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What on Earth is AlphaEarth? Exploring functional specialization and shared associations in Geospatial Embeddings for Land-Cover analysis

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***Abstract*—Geospatial foundation models generate high-dimensional embeddings that achieve strong predictive performance, yet their internal organization remains obscure, limiting their scientific use. Recent interpretability studies relate Google AlphaEarth Foundations (GAEF) embeddings to continuous environmental variables, but it is still unclear whether the embedding space exhibits a functional or hierarchical organization, in which some dimensions act as specialized representations while others encode shared or broader geospatial structure. We present a post hoc framework for exploring the functional roles of pretrained AlphaEarth Foundations embedding dimensions in land-cover discrimination. The framework combines geographically distributed one-vs-rest classification experiments, model-derived feature rankings, progressive subset retraining, and class–dimension association analysis. It characterizes dimensions according to the breadth of their associations with land-cover classes, distinguishing class-specific and shared roles within the evaluated tasks. These categories provide an empirical basis for interpreting recurring discriminative patterns and formulating hypotheses about their environmental meaning. Using a 98% relative baseline-recovery criterion across eight classification metrics, the analysis selected between 2 and 12 dimensions per land-cover class from experiment-specific feature rankings. Aggregate top-two frequencies were then used to construct class-level interpretive sets, whose predictive performance as fixed subsets remains to be evaluated. The results indicate opportunities for task-specific dimensionality reduction and provide an exploratory taxonomy of class–dimension associations. The framework supports the formulation of testable interpretations of pretrained geospatial representations; the physical meaning of individual dimensions and the transferability of the proposed associations require further validation.**

***Index Terms*—Geospatial Foundation Models, Earth Observation, Remote Sensing, Virtual Satellites, Explainable AI, Responsible AI, Feature Attribution, Interpretable Embeddings.**

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I. Introduction

GEOSPATIAL foundation models (GFMs) — part of a broader movement toward comprehensive world models that simulate the dynamics of Earth's interconnected systems — are transforming Earth observation research in fields such as ecology, infrastructure planning, and climate science [3, 5]. These models function as "virtual satellites" capable of characterizing the Earth's surface and its dynamics with an unprecedented level of detail [6]. GFMs have emerged as a response to two classic bottlenecks in Earth observation: converting large volumes of multi-source data into useful information and the limited generalization of task-specific models when they depend on high-quality labels [5]. In this context, Google AlphaEarth Foundations (GAEF) was proposed as a task-agnostic learned featurization approach that integrates multiple data sources, producing eight annual 64-dimensional embeddings at 10 m (2017 to 2024) to support operational mapping and learning with sparse labels [4, 5].

These developments build on advances in spatial–spectral sensing and analysis. Hong et al. [9] review the physical principles, acquisition and calibration procedures, and analytical methods underlying hyperspectral imaging, including dimensionality reduction, classification, spectral unmixing, and emerging artificial intelligence approaches. This perspective places learned representations within a broader measurement-to-analysis workflow. For the present study, an important distinction is that sensor bands correspond to spectral measurements, whereas embedding dimensions are learned coordinates whose environmental meaning requires empirical investigation.

Recent architectures address spatial–spectral learning through different objectives. SpectralGPT [10] uses a three-dimensional generative pretrained transformer with spatial–spectral tokenization and reconstruction-based learning to support downstream classification, semantic segmentation, and change detection. MambaX [13] develops a nonlinear state predictive control

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approach for image super-resolution and multimodal fusion, mapping consecutive spectral bands into a latent state space. These studies illustrate advances in representation learning and image reconstruction that provide context for examining the functional behavior of pretrained geospatial embeddings.

GAEF addresses traditional limitations of remote sensing, such as source heterogeneity, differences in spatial and temporal resolution, and the difficulty of integrating multiple sensors by synthesizing large volumes of data from optical images, radar, LiDAR, climate variables, and other sources (Figure 1 left) into a unified and consistent representation of the planet [11, 6]. However, despite demonstrated effectiveness in various mapping and environmental modeling tasks [4, 15], these models present a fundamental interpretability challenge: they encode environmental information into high-dimensional embeddings that compress complex geophysical patterns into abstract numerical vectors (Figure 1 right) [4, 11]. Several criticisms have recently been raised about GAEF, arguing that, despite its strong benchmark performance, it functions as an opaque, annually updated 10-m "black-box" representation whose mixed-pixel embeddings, limited physical and human-centric interpretability, and dependence on biased upstream data constrain its reliability for high-stakes, fine-scale, and socio-ecological decision-making in real-world settings [17, 14].

Unlike conventional remote sensing products, where each band is associated with a measurable physical property such as spectral reflectance, temperature, or humidity [18], these embeddings represent latent features that capture complex Earth surface patterns without an explicit biogeophysical interpretation [14]. Their values, usually defined in normalized, high-dimensional spaces, do not offer a direct interpretation of the environmental processes they represent, creating a significant gap between the predictive capacity of the models and their scientific understanding [17, 14]. This limitation raises the need to develop methodological frameworks that allow these embeddings to be interpreted in terms of trackable and reproducible physical variables and environmental processes, facilitating their integration into scientific decision-making. Without such frameworks, the utility of embeddings is limited to predictive performance, with little capacity to explain what environmental information is encoded or how they relate to the biogeophysical characteristics of the Earth's surface. Bridging this gap is particularly relevant for applied domains such as climate adaptation, ecological monitoring, and infrastructure planning, where understanding the structure of model outputs can inform more targeted and efficient use of these representations.

Furthermore, applied studies explicitly report limitations of GAEF in contexts where interpretability and time sensitivity matter (e.g., agriculture), pointing to low interpretability and other practical constraints such as spatial transferability and limited time sensitivity [15]. Evidence suggests that GAEF content is rich but poorly characterized: its geometric properties (e.g., normalized embeddings on the unit hypersphere) and its use as a consistent global basis are known, but there is no operational consensus on which dimensions correspond to which variables/processes, nor on how to robustly validate these associations [14, 17].

The objectives of representation learning, image reconstruction, and post hoc interpretation are related but distinct. SpectralGPT evaluates the downstream utility of learned spectral representations [10], whereas MambaX investigates reconstruction through state-space modeling [13]. Our study addresses a complementary question: how dimensions of a released, pretrained representation participate in land-cover discrimination, and how their class-specific and shared associations recur across experimental contexts. This distinction motivates examining functional associations alongside predictive performance, without assuming that a useful latent coordinate has a unique physical interpretation.

Recent studies have begun to show that GAEF embeddings contain physically meaningful information that extends well beyond their original mapping objectives. [17] demonstrated that individual embedding dimensions can be associated with continuous environmental variables — such as temperature, vegetation indices, hydrology, and terrain — and that these associations enable large language models to generate interpretable descriptions of land surface conditions with high accuracy. [16] showed that GAEF embeddings provide richer and more transferable representations of basin characteristics than hand-crafted hydrological attributes, improving streamflow prediction in ungauged basins by capturing integrated environmental signals related to topography, vegetation, and soil properties. [12] further demonstrated that GAEF embeddings constitute effective covariates for modelling soil organic carbon and water table depth in peatlands, performing comparably to expert-derived remote sensing inputs while offering substantially lower data preparation barriers.

These AlphaEarth-specific studies examine physical associations and downstream predictive utility. A complementary question is how individual embedding dimensions contribute to discriminating categorical land-cover classes, and whether these contributions recur across geographically distributed experiments. We address this question by characterizing the breadth and spatial consistency of class–dimension associations. Land cover provides a defined task domain for this analysis, allowing functional roles to be examined relative to a common reference legend. The resulting interpretation is conditional on that domain, the reference labels, and the downstream classifiers.

We investigate whether pretrained AlphaEarth Foundations dimensions show recurring differences in their contribution to land-cover discrimination across sampled geographic contexts. Here, functional interpretability refers to characterizing a dimension through its observed role in downstream classification tasks. We examine class-specific and shared associations and use them to formulate environmental hypotheses. The analysis operates on released embeddings without modifying the foundation model.

The contribution is an integrated workflow that connects experimental feature rankings and reduced-dimension performance profiles with class-level association summaries, an operational specialist–generalist taxonomy, and spatial and interactive exploration. The individual classifiers and feature-importance measures are established methods; the methodolog-

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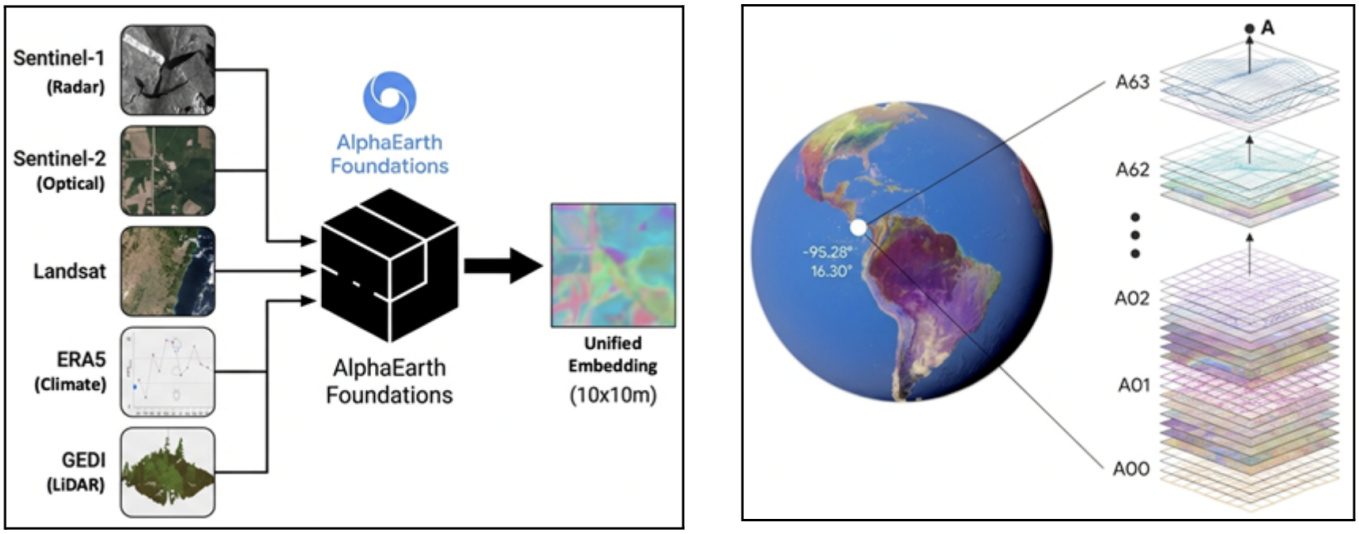


Fig. 1: Multi-modal fusion at scale, left) data containing the features, and right) its representation in an embedding.

ical contribution lies in their combination to characterize the functional behavior of pretrained geospatial dimensions. The empirical evaluation concerns AlphaEarth Foundations and the ESA WorldCover 2020 legend.

II. Materials and Methods

A. Overall study design

This study proposes a functional interpretability framework for GAEF based on a combination of massive experimentation and structural analysis of the embedding space. The approach is based on the use of embeddings generated by GAEF, which are high-dimensional representations that integrate spatial, temporal, and multi-modal information about the Earth's surface into a unified latent space [4].

The proposed methodology is structured around two complementary components: a massive experimental exploration aimed at empirically characterizing the discriminative behavior of embeddings; and a structural analysis based on feature importance patterns and progressive ablation that allows interpretations of the functional organization of the representation space in terms of land cover types and their interactions. These components are visualized with an interactive Dashboard that enables exploratory analysis of the embedding space structure and its relationship to land cover classes [8].

The classification experiments serve as probes of the released representation. Geographic distribution across multiple continents broadens the contexts examined, while the evaluation remains a collection of local one-vs-rest tasks. Accordingly, "global" describes the geographic scope of the experimental sampling. The design does not establish exhaustive global mapping accuracy, spatially independent transfer performance, or generalization across years and foundation models (Figure 2).

B. Data and representations

The analysis is based on the integration of two main sources of information: land cover labels and the 64 GAEF embeddings. The labels come from the European Space Agency's (ESA) WorldCover 2020 product, which provides a global classification of the Earth's surface into 11 discrete categories [19]. ESA WorldCover was selected for several reasons: its land cover classes are globally generalizable, providing a consistent categorical framework across diverse geographic contexts; its high spatial resolution (10 m) supports fine-grained discrimination between classes; and its temporal alignment with the period over which GAEF was constructed ensures that the land cover labels correspond to the same conditions encoded in the embeddings. Crucially, ESA WorldCover is independent from the training data used in the construction of GAEF embeddings, meaning that the classification experiments evaluate the embeddings against an external reference rather than recovering labels already encoded during training. These classes represent the categorical structure of the Earth's surface and constitute the target variable in the classification experiments.

The explanatory variables used are the GAEF embeddings, which correspond to the 64-dimensional vectors produced from the integration of multiple observation sources, including optical sensors, radar, climate variables, and other Earth observation products [4, 11]. Each observation was represented by a 64-dimensional AlphaEarth Foundations embedding. The original Earth Engine band identifiers, A00–A63, are used throughout the manuscript. These variables represent coordinates of a learned embedding space rather than physical spectral bands.

C. Massive experimental exploration of embeddings

The first methodological component consists of a large-scale automated experiment designed to evaluate the discriminatory capacity of embeddings in land cover classification tasks. This experiment involves the execution of 110,000 independent analyses, each structured as a binary classification in which a target land cover class is evaluated against an aggregation of all remaining 10 classes, thereby isolating the embedding dimensions most informative for distinguishing each individual

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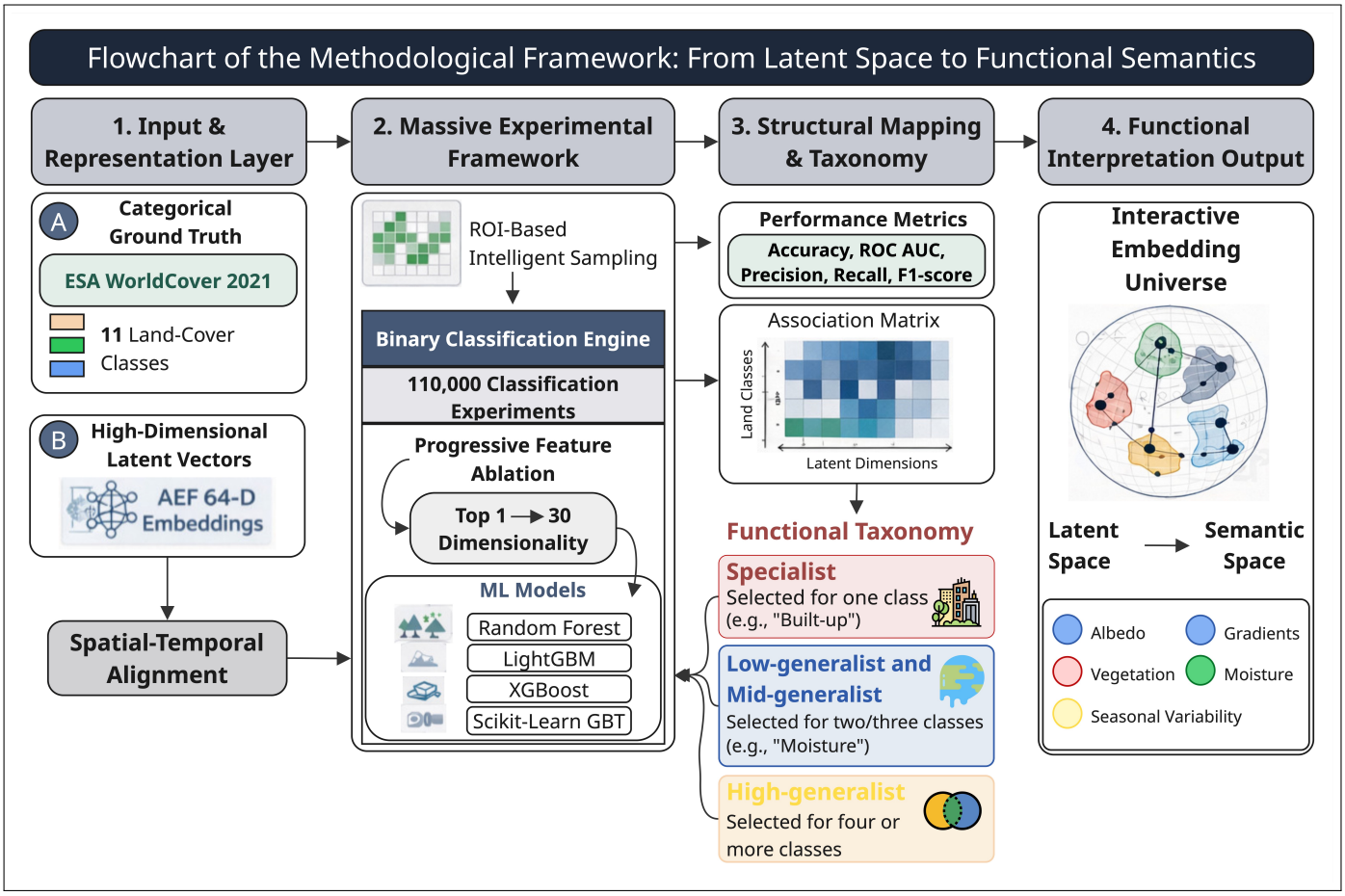


Fig. 2: Flowchart of the methodological framework: from embeddings to functional semantics. The pipeline integrates (1) ESA WorldCover 2020 categorical labels and AEF 64-dimensional embeddings as inputs, (2) a massive experimental framework executing 110,000 binary classification experiments using Random Forest, Gradient Boosting, XGBoost, and LightGBM with progressive feature ablation across the top 1 to 30 embedding dimensions, (3) structural mapping via an association matrix and functional taxonomy, and (4) an interactive embedding universe visualization connecting the latent space to semantic space.

land cover type. To explore the fundamentals of experimental design and run experiments independently, refer to the GAEF Land Cover Classification App and to the Python pipeline developed to run the full set of experiments [1, 2].

To ensure that each experiment contains sufficient representation of the target class, a class-presence-guided spatial selection procedure is implemented using Google Earth Engine. For each analysis, a continent is randomly selected from a global set and the ESA WorldCover layer is queried to identify a location where the target land cover class is present. Then a rectangular region of interest (ROI) of randomized dimensions $(0.1 - 1.0)$ is constructed around this location. This approach ensures that each ROI contains pixels of the target class while preserving spatial variability across experiments so that the global diversity of land cover conditions is represented. In cases where no target class pixels are found within the selected continent, the system falls back to a random ROI within the continental bounds.

Once the analysis region has been defined, the land cover labels and embedding dimensions corresponding to the selected pixels are extracted simultaneously via stratified sampling, ensuring balanced representation of the target class and all other classes. These data are used to build training and validation sets, in a split of $75/25$, allowing the discriminative ability of each embedding to be evaluated between classes. For each experiment, one of four machine learning algorithms is randomly selected: Random Forest, Scikit-Learn's Gradient Boosted Trees, XGBoost, and LightGBM. All four are suitable for capturing nonlinear relationships and have proven effective in remote sensing classification tasks.

Each experiment follows a training and evaluation scheme in which a model is first trained using all 64 embedding dimensions, and the relative contribution of each dimension to the classification is estimated using the native feature importance scores of the trained model, based on Mean Decrease in Impurity (MDI) across decision tree splits. Based on this ranking, a progressive ablation procedure is implemented in which models are sequentially retrained using subsets of the embedding dimensions, incrementally expanding from the single most important dimension up to the top 30, as ordered by their MDI-based importance scores. This approach allows for the analysis of the redundancy and complementarity of

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the information contained in the embeddings, as well as the evaluation of how many dimensions are necessary to achieve adequate performance. Following the initial filtering, 10,000 records per land-cover class were randomly selected using seed 123, yielding 110,000 records. The performance-curve analysis excluded 43 selected records with missing Top-1 accuracy, leaving 109,957 records. All baseline and Top-1 to Top-30 metric values were available for the retained records. For each land-cover class, performance was summarized across experiments separately for eight metrics: accuracy, ROC AUC, and precision, recall, and F1-score for each binary class. For each metric, we identified the first evaluated subset size, from 1 to 30 dimensions, whose mean score reached at least 98% of the corresponding mean score obtained with all 64 dimensions. The class-level dimension count was defined as the maximum of these eight first-crossing sizes. When several metrics shared this maximum, the summary recorded the first one in the evaluation order as the limiting metric. The 98% threshold denotes relative recovery of baseline performance, rather than an absolute classification accuracy of 98%. The reduced models used experiment-specific feature rankings, so the identities of the retained dimensions could differ between experiments. We additionally verified that, at the selected class-level dimension count, all eight mean metric scores simultaneously reached at least 98% of their corresponding 64-dimension baselines. For every class, this count also coincided with the first evaluated subset size satisfying all eight criteria simultaneously. All metrics remained above their respective thresholds through the largest evaluated subset of 30 dimensions, despite small fluctuations in the mean performance curves.

The results of each experiment — including performance metrics, variable importance, geographic location, and algorithm used — are systematically recorded in a structured database. This database constitutes the main input for subsequent interpretive analysis, providing large-scale empirical evidence on the behavior of these embeddings.

D. Structural analysis of the embedding space

The second methodological component focuses on interpreting the embedding space based on the experimental results. While the previous phase allowed us to identify the relevance of each dimension in specific tasks, this phase seeks to characterize the overall organization of the representation space in relation to land cover types.

To do this, we construct an association matrix that quantifies the relationship between each embedding dimension and the different land cover classes (Table I). For each land cover class and each embedding dimension, the association score is computed as the normalized frequency with which that dimension ranked among the two most important features by MDI across all experiments targeting that class. Formally, the association score for a given class–dimension pair is defined as the number of experiments in which the dimension appeared among the top two MDI-ranked features, divided by the total number of experiments for that class, yielding a value between 0 and 1. This normalization ensures comparability across classes with different numbers of experiments. The resulting matrix provides a quantitative summary of the discriminative relevance of each embedding dimension for each land cover class, and serves as the basis for the fingerprint plot (Figure 3), the conceptual embedding space representation (Figure 4), and the embedding universe visualization (Figure 6) implemented in the interactive Dashboard [8].

The class-level interpretive sets were constructed by combining the dimension counts obtained in Section II.C with the association scores defined above. For each land-cover class, dimensions were ranked by their frequency of occurrence among the two most important features across experiments targeting that class. The highest-ranked dimensions were retained up to the corresponding class-level dimension count. Thus, the performance analysis determined how many dimensions to select, whereas the aggregate frequency ranking determined which dimensions entered the interpretive set. These fixed class-level sets were not evaluated by retraining classifiers with their selected dimensions.

A dimension was designated a specialist if it was selected for one land-cover class. Dimensions selected for two, three, or four or more classes were designated low-, mid-, and high-generalists, respectively. All non-specialist selected dimensions were collectively referred to as shared dimensions. Dimensions not selected for any class remained uninterpreted under this procedure.

The specialist–generalist categories are operational descriptors of association breadth within the selected class-level sets. “Specialist” denotes a dimension selected for one class, whereas the generalist categories denote selection for progressively more classes. This ordering summarizes the observed association structure; it does not establish an intrinsic hierarchy of abstraction within AlphaEarth Foundations. The assigned roles depend on the reference classes, experimental design, feature-ranking procedure, and subset-selection rule.

Additionally, Figure 3 presents what we term the “Embedding Fingerprint” plot — a visualization whose structure deliberately echoes that of a genomic fingerprint, where the presence and absence of band patterns at specific loci encode the identity of a biological sample. In our case, the on/off pattern of exclusive (blue) and shared (pink) embedding dimensions across positions A00–A63 encodes the functional identity of each land cover class, much as specific band combinations in a DNA fingerprint encode the identity of an organism. Just as no two species share an identical banding pattern, each land cover class exhibits a distinctive dimensional signature: specialist dimensions act as unique genetic markers exclusive to a single class, while shared dimensions — encompassing low-, mid-, and high-generalists — function as conserved sequences that appear across multiple classes, reflecting structural or spectral similarities between them. This visualization summarizes the composition of the class-level interpretive sets and the distribution of class-specific and shared associations. Set sizes are derived from the 98% relative performance-recovery procedure, whereas dimension identities are determined by aggregate top-two frequencies. The visualization does not establish the predictive performance of these fixed sets.

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TABLE I: Excerpt of the association matrix quantifying the normalized frequency with which each embedding dimension ranked among the two most important features by MDI across all experiments targeting each land cover class. Values range from 0 to 1, where higher values indicate stronger discriminative relevance of that dimension for the corresponding class

	impA00	**impA01**	**impA02**	**impA03**	···	**impA63**
Bare/sparse vegetation	0.0267	0.0223	0.0350	0.0167	···	0.0409
Built-up	0.0044	0.1092	0.0478	0.0057	···	0.0040
Cropland	0.0163	0.0266	0.0564	0.0369	···	0.0179
Grassland	0.0211	0.0315	0.0552	0.0270	···	0.0161
Herbaceous wetland	0.0177	0.0449	0.0269	0.1090	···	0.0199
Mangroves	0.0446	0.0103	0.0112	0.0162	···	0.0093
Moss/lichen	0.0093	0.0265	0.0338	0.0136	···	0.0285
Shrubland	0.0140	0.0278	0.0892	0.0178	···	0.0190
Snow/ice	0.0043	0.0041	0.1583	0.0778	···	0.0063
Tree cover	0.0314	0.0208	0.0454	0.0248	···	0.0315
Permanent water bodies (”Water”)	0.0381	0.0983	0.0659	0.0180	···	0.1386

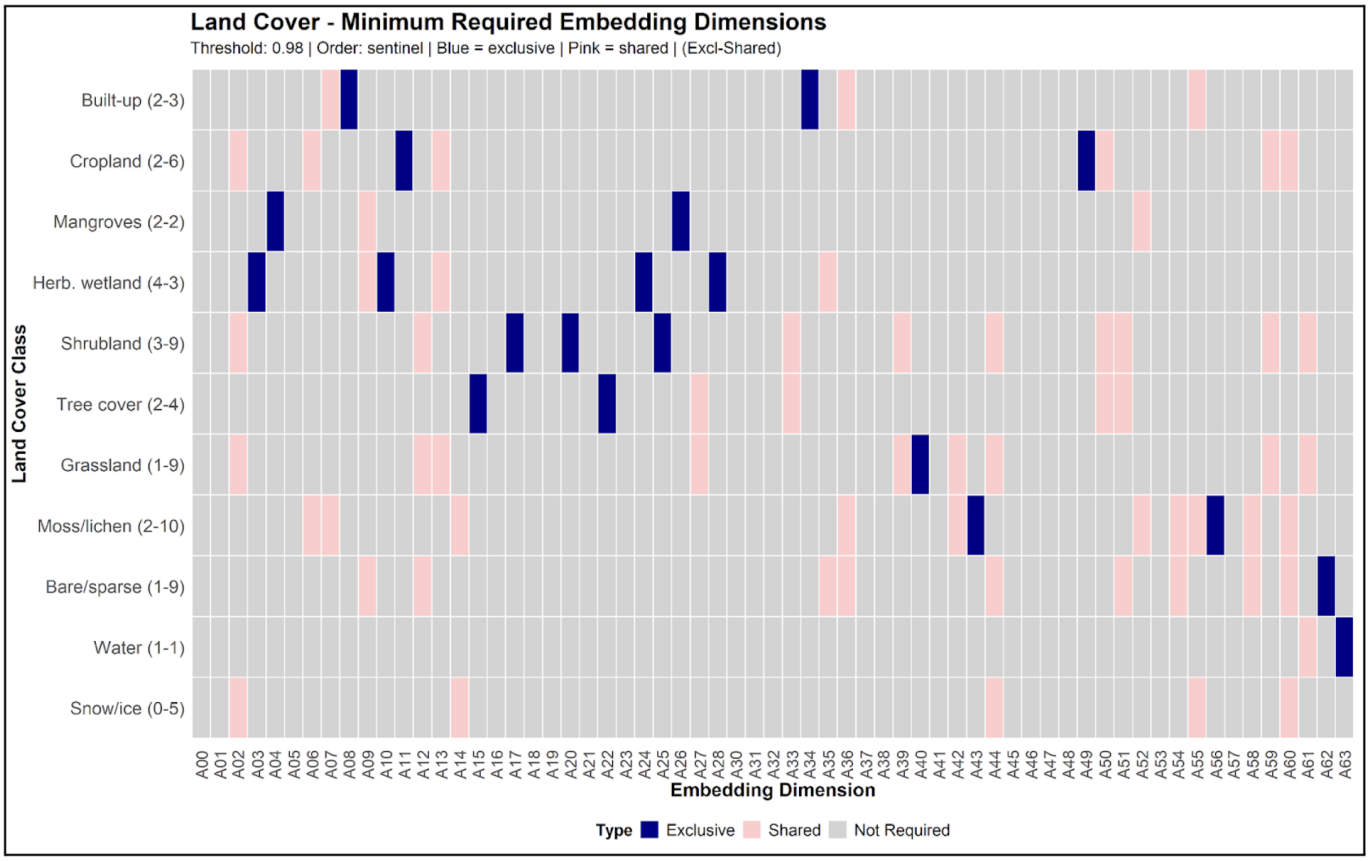


Fig. 3: *Embedding Fingerprint* plot illustrating the functional dimensional signature of each land cover class at the 98% performance recovery threshold. Analogous to a genomic fingerprint, where the presence and absence of bands at specific loci encodes biological identity, each row represents the characteristic pattern of exclusive (blue) and shared (pink) embedding dimensions that together define the discriminative identity of a land cover class within the AEF embedding space. Exclusive dimensions appear in only one class's minimum subset, while shared dimensions — encompassing low-, mid-, and high-generalists — contribute to the classification of two or more classes. Gray positions indicate dimensions not required to achieve the 98% threshold for that class. Land cover classes are ordered from bottom to top by increasing spectral complexity, quantified as the mean pixel-wise variance across Sentinel-2 spectral bands computed over 100 randomly selected images per class after standard atmospheric and radiometric corrections. Spectrally homogeneous classes such as Snow/ice and Water — which exhibit low inter-pixel variability due to their uniform reflectance signatures — appear at the bottom, while spectrally heterogeneous classes such as Built-up — whose radiometric response varies substantially across rooftops, roads, and impervious surfaces — appear at the top, reflecting the increasing number of embedding dimensions required to characterize them.

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E. Spatial consistency of interpreted dimensions

The association scores described above aggregate over all experiments for a given class and therefore describe the *global-average* behavior of each embedding dimension. To assess whether these interpretations hold uniformly across the Earth's surface or instead reflect geographically localized behavior, we evaluated the spatial consistency of each interpreted dimension. Using the centroid coordinates of each experiment's region of interest, experiments were binned into a $50° \times 50°$ longitude $\times$ latitude grid, and for every dimension–class pair we computed the top-two frequency within each grid cell, i.e. the proportion of that cell's experiments in which the dimension ranked among the two most important features by MDI. To avoid over-interpreting sparsely sampled regions, cells containing fewer than a minimum number of experiments (15) were excluded, and dimensions retaining fewer than three reliable cells were not scored.

For each dimension–class pair we summarized the resulting field of per-cell frequencies into a single spatial-consistency index bounded on $[0, 1]$, where values near 1 indicate a dimension that discriminates its associated class uniformly across space and values near 0 indicate strongly localized, region-specific behavior. The index combines the spatial uniformity of the frequency field, quantified as one minus its N-weighted coefficient of variation, with its relative discriminative strength, so that a dimension is scored highly only when it is both evenly distributed and consistently informative; complementary uniformity measures (N-weighted Pielou evenness and the fraction of cells exceeding a presence threshold) were computed alongside to distinguish low scores arising from spatial spikiness, weak overall importance, or narrow spatial extent.

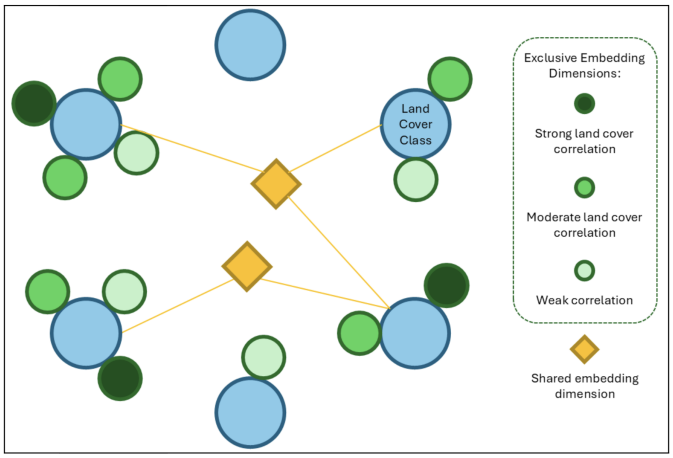


Fig. 4: Conceptual structure of the embedding space representation, illustrating the planet-moon organization of the embedding universe. Blue circles represent land cover classes (planets), while green circles represent exclusive embedding dimensions (moons) whose color intensity encodes association strength — dark green indicating strong discriminative relevance, medium green indicating moderate relevance, and light green indicating weak relevance for that class. Gold diamonds represent shared embedding dimensions that connect multiple land cover classes, positioned at the geometric centroid of their associated classes. This conceptual layout serves as the interpretive basis for the embedding universe visualization implemented in the interactive dashboard [8] and shown in Figure 6. Node positions are assigned for visualization and do not represent measured distances in the original embedding space or geographic locations of ecological transitions.

F. Visualization and interpretation of the universe of embeddings

To facilitate interpretation of the embedding space structure, we developed the *"What on Earth is AlphaEarth?"* interactive dashboard, which visualizes the relationships between embedding dimensions and land cover classes in a unified graphical environment [8]. The dashboard integrates classification outputs, embedding importance metrics, and geographic context, enabling users to explore the functional organization of the embedding space (Figures 4 and 6). In its primary view, land cover classes are organized as main nodes arranged in a circular space, while embedding dimensions are represented as entities associated with these classes.

Embedding dimensions classified as specialists are visualized as elements close to a single coverage, reflecting their exclusive association. The darkness of each embedding dimension represents the strength of its association with a particular land cover class, where a darker green indicates a stronger association and a lighter green indicates a weaker one. In contrast, shared embedding dimensions — those associated with multiple land cover classes — are located in intermediate positions between their associated coverages, reflecting their role in capturing characteristics common to more than one class. Notably, such shared dimensions may also encode information relevant to ecological transition zones, where land cover types co-occur with one another. This spatial arrangement is calculated using geometric centroids, positioning each embedding dimension according to its relationships with the land cover classes it characterizes.

The Dashboard allows intuitive exploration of the structure of the embedding space, identifying patterns of specialization, clustering, and connectivity between land cover classes. This approach facilitates the interpretation of complex results and provides visual evidence of the functional organization of the embeddings.

The Dashboard comprises four integrated views. The Overview provides a synthesis of the global experiments, including model performance comparisons across the four algorithms employed and a global map of experiment locations. The Class Analysis view houses the "Embedding Universe" visualization (Section 3.6), class performance matrices (Section 3.2), embedding importance charts that enable direct comparison of dimensional contributions across classes (Sections 3.2, 3.3). The Geographic view maps individual experiment bounding boxes onto satellite imagery alongside spatially aggregated performance heatmaps, enabling identification of geographic regions where the embedding space performs strongest or weakest (Supplementary Figure 2). Finally, a Chat interface serves as a companion to the experimental notebook

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(Section 2.3), guiding users through the configuration of new experiments.

G. Interpretive synthesis and hypothesis generation

Interpretation proceeds from observed class–dimension associations to candidate environmental explanations. For dimensions selected for a single class, we consider properties characteristic of that class as possible explanations for their discriminative relevance. For shared dimensions, we examine whether the associated classes have plausible spectral, structural, phenological, or geographic relationships. These explanations organize the observed patterns and identify hypotheses for subsequent testing.

Association with a class does not establish that a dimension directly measures a particular physical property. Likewise, selection across several classes does not establish sensitivity to their spatial boundaries or interactions. The biophysical descriptions in Tables II and III are therefore proposed interpretations rather than validated semantic assignments.

III. Results

A. Empirical evidence of the informational efficiency of embeddings

The results of the large-scale experimental exploration reveal that the discriminatory capacity of GAEF embeddings is not uniformly distributed in the representation space. A small subset of dimensions concentrates information necessary to differentiate land cover types, and models trained with a limited number of prioritized dimensions achieve performance levels comparable to those obtained with the full set of 64 embeddings (Figure 5a,b).

Across the 11 ESA WorldCover classes, the class-level counts obtained from the eight metric-specific first crossings ranged from 2 to 12 dimensions (Figure 5c). The selected counts were 2 for Water, 4 for Mangroves, 5 for Built-up and Snow/ice, 6 for Tree cover, 7 for Herbaceous wetland, 8 for Cropland, 10 for Bare/sparse vegetation and Grassland, and 12 for Moss/lichen and Shrubland. These counts summarize the performance curves obtained with experiment-specific feature rankings; they do not constitute a predictive evaluation of the fixed class-level interpretive sets.

In all cases, performance curves exhibit a characteristic plateau, with rapid gains from the first few dimensions followed by diminishing returns as additional dimensions are added (Supplementary Figure 1). This pattern suggests the existence of considerable redundancy in the embedding space alongside a core of highly informative dimensions that capture distinctive characteristics of each land cover type. The variation in tipping points across classes further indicates that the embedding space encodes class-specific information at different levels of concentration — classes with unique spectral or spatial signatures are captured by fewer, more specialized dimensions, while classes sharing characteristics with others require a broader set. This finding is consistent with the nature of representations learned in foundation models, where information tends to be organized into distributed but non-homogeneous latent structures [17].

The reduced feature counts motivate further evaluation of computational efficiency. The magnitude of any savings in training time, inference time, or memory use must be established through direct measurements for the algorithms and deployment settings considered.

B. Emergence of highly specialized embedding dimensions

The analysis of the importance of variables reveals the recurring presence of embedding dimensions with a high contribution to the discrimination of specific coverages. These dimensions show a strong association with a single class, maintaining limited relevance in other categories.

This pattern suggests the existence of highly specialized embedding dimensions that show recurring discriminative relevance of certain land cover types. In particular, it is observed that land cover types with clearly distinguishable physical signatures, such as bodies of water or urban areas, tend to have embedding dimensions with a high concentration of importance.

The emergence of these specialized embedding dimensions is consistent with the ability of foundation models to encode physical and environmental properties in high-dimensional latent spaces [17]. However, this work does not seek to directly associate embedding dimensions with specific physical variables such as temperature or elevation. Instead, it interprets their behavior functionally, examining how individual dimensions contribute to discriminating between land cover classes and capturing spatial interactions among them.

C. Evidence of shared embedding dimensions between coverages

Additionally, the analysis shows the existence of embedding dimensions that make significant contributions across multiple coverages. These dimensions are not exclusively associated with one class, but rather participate in the discrimination of different types of coverage. This behavior suggests that certain embedding dimensions capture common patterns or interactions between land cover classes, which is particularly relevant in contexts where classes are not strictly separable. From a geographical perspective, this type of representation is consistent with the existence of transition zones or ecotones, where characteristics of multiple land coverages coexist. From a functional perspective, it is also consistent with land cover classes that share spectral, phenological, or structural characteristics, such as croplands and grasslands, which may exhibit similar seasonal greenness patterns despite differing in management and land use.

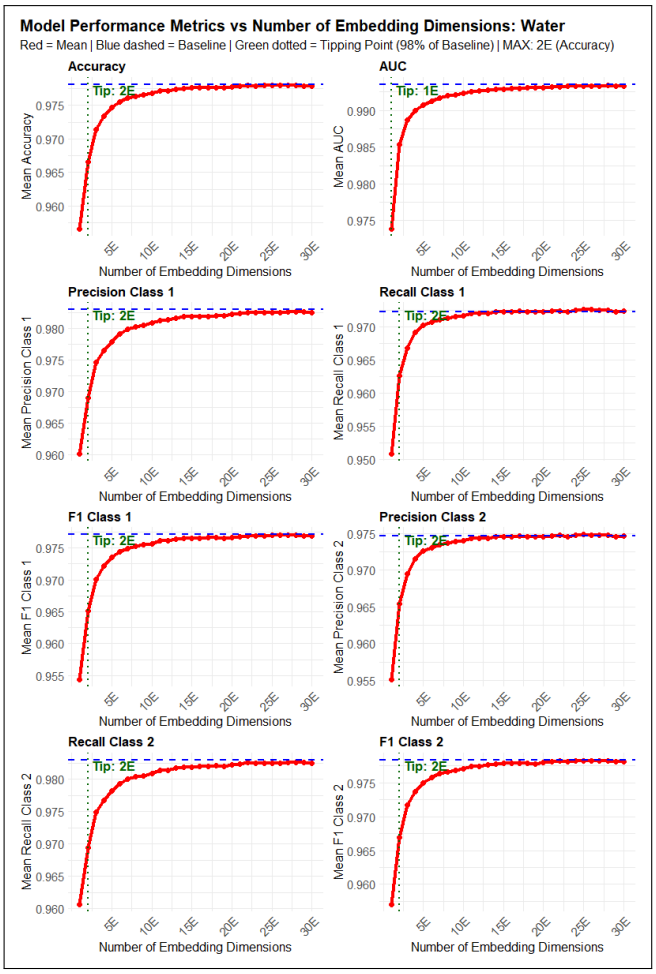
The presence of these shared embeddings is consistent with the ability of GFMs to integrate multi-source information and capture complex spatial dynamics [4]. In this sense, these dimensions cannot be interpreted as exclusive attributes, but rather as components that reflect relationships between classes.

D. High-generalist embedding dimensions and global behavior

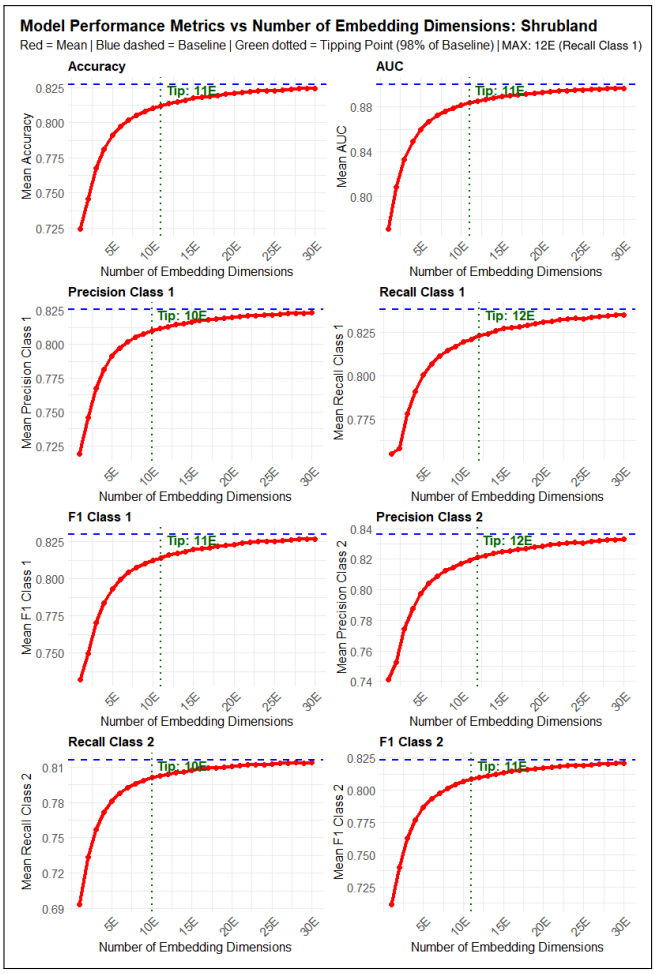
Among the shared dimensions reported in Table III, high-generalist dimensions (those associated with four or more

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(a)



(b)

Land Cover Class	Number of Embedding Dimensions
Water	2
Mangroves	4
Built-up	5
Snow/ice	5
Tree cover	6
Herbaceous wetland	7
Cropland	8
Bare/sparse vegetation	10
Grassland	10
Moss/lichen	12
Shrubland	12

(c)

Fig. 5: Performance of progressively expanded feature subsets and selected dimension counts by land-cover class. Panels (a) and (b) show eight classification metrics for Water and Shrubland, respectively, using subsets selected from experiment-specific feature-importance rankings. Red curves represent mean performance across experiments, blue dashed lines indicate the corresponding 64-dimension baselines, and green dotted lines mark the metric-specific first crossings of 98% of baseline performance. Panel (c) reports the maximum of these eight first-crossing sizes for each of the 11 ESA WorldCover classes. These counts range from 2 dimensions for Water to 12 for Moss/lichen and Shrubland.

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TABLE II: Exclusive embedding dimensions and their associated ESA WorldCover 2020 land cover classes, listing dimensions selected exclusively for one class-level interpretive set alongside a biophysical interpretation of the distinctive spectral, structural, or phenological properties that justify their exclusive association.

Exclusive Embedding Dimension	**Associated Land Cover Class**	**Candidate biophysical explanation for the observed association**
A08, A34	Built-up	Urban surfaces and infrastructure: roofs, roads, concrete. Their radiometric signature is dominated by artificial materials with high SWIR reflectance compared to NIR and strong Sentinel-1 backscatter. No other class combines this angular geometry and artificial-material spectrum, which justifies specific embeddings.
A11, A49	Cropland	Annual crops sown and harvested within a 12-month cycle, with very regular phenology, geometric patterns and often intensive management. Although they can be confused with grasslands, certain regions of the feature space carry signatures specific to cropland, explaining their exclusive embeddings.
A04, A26	Mangroves	Evergreen woody vegetation tolerant to salinity in the intertidal zone. They show dense canopy like Tree cover, but with coastal linear patterns, water mixed between crowns and specific NIR/SWIR signatures due to permanent waterlogging. This combination is unique to mangroves.
A03, A10, A24, A28	Herbaceous wetland	Areas dominated by herbaceous vegetation that is permanently or regularly flooded. They combine moderate–high NDVI with very high water indices and floodplain textures; no other type combines herbaceous cover and long-lasting flooding in this way.
A17, A20, A25	Shrubland	Zones dominated by natural shrubs, typical of semi-arid biomes. Although they mix with grassland and bare/sparse vegetation, some subtypes generate signatures distinct enough to occupy an exclusive subspace.
A15, A22	Tree cover	Regions dominated by trees, including plantations, with very high NIR, strong shadows and closed canopy textures. This 3-D structure and phenological persistence clearly separate them from shrubland and grassland.
A40	Grassland	Areas dominated by natural herbaceous vegetation with marked seasonality but homogeneous structure, without long-lasting flooding or intense geometric traces. Embedding A40 likely captures well-defined grassland far from crops and shrubs.
A43, A56	Moss/lichen	Surfaces dominated by mosses and/or lichen, typical of tundra and cold or very arid rocky environments. Their low structure and spectral behaviour differ from tall herbaceous vegetation and shrubs.
A62	Bare/sparse vegetation	Lands with soil, sand or exposed rock and less than 10% vegetation throughout the year. This is the purest bare soil signature, with high VIS and SWIR reflectance, minimal chlorophyll signal and little temporal variation.
A63	Permanent water bodies	Permanent water bodies show very strong absorption in NIR and SWIR, clear linear or polygonal geometry and low spectral variability. This near-zero NIR/SWIR reflectance and smooth texture is unique compared to other classes.

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TABLE III: Shared embedding dimensions and their functional interpretation, according to the number of class-level interpretive sets in which they were selected. For each dimension, the associated common ecological–spectral trait are reported, inferred through the two-stage interpretive process described in Section 2.6.

Embedding Dimension	Associated ESA Land Covers	Categorization	Proposed interpretive label	Candidate ecological or spectral explanation
A06	Cropland, Moss/lichen	Low-generalist	Discontinuous vegetation	Areas with low and discontinuous vegetation over mineral soil: croplands in fallow or early phenological stages and lichen mats show a soil and sparse vegetation mixture with high visible albedo and marked seasonal variability.
A07	Built-up, Moss/lichen	Low-generalist	Sparse vegetation over bright surfaces	Bright, sparsely vegetated substrates such as roofs, concrete and rock with lichens share low chlorophyll signal and rough textures that the model groups as mineral surfaces with minimal vegetation.
A14	Moss/lichen, Snow/ice	Low-generalist	High latitude and high albedo	High-latitude or high-altitude cold environments where snow/ice and lichen mats over rock produce bright visible reflectance and spatially homogeneous cryogenic landscapes.
A27	Grassland, Tree cover	Low-generalist	Dense green vegetation	Landscapes dominated by green vegetation with strong NIR response where the main difference is canopy density; the embedding captures forest–grassland mosaics with intermediate canopy fraction.
A33	Shrubland, Tree cover	Low-generalist	Woody perennial	Woody perennial vegetation with relatively high biomass; shrublands and forests differ in height but share canopy textures and similar spectral and SAR responses.
A35	Bare/sparse, Herbaceous wetland	Low-generalist	Soil-moisture gradient	Transition from dry bare soil to wet herbaceous substrates where exposed mud and seasonal wetlands resemble moist bare surfaces, capturing gradients of soil moisture.
A39	Grassland, Shrubland	Low-generalist	Open low–medium vegetation	Continuum of open vegetation of low to medium height typical of grasslands and shrublands that share discontinuous structure and similar spectral behaviour.
A42	Grassland, Moss/lichen	Low-generalist	Low-height vegetation	Surfaces with creeping or very low vegetation such as sparse grasslands or lichen-dominated tundras, producing moderate NDVI and frequent mixing with exposed soil.
A52	Mangroves, Moss/lichen	Low-generalist	Extreme conditions	Vegetation adapted to extreme environments: saline mangroves and cold or arid lichens both show specialized perennial vegetation and complex textures linked to environmental stress.
A54	Bare/sparse, Moss/lichen	Low-generalist	Rocky sparse vegetation	Mineral substrates dominated by rock or soil with extremely sparse vegetation where lichen patches barely increase NDVI above bare soil levels.
A58	Bare/sparse, Moss/lichen	Low-generalist	High-reflectance sparse vegetation	Reinforces the embedding subspace associated with high-reflectance surfaces with very low vegetation cover where lichens and bare soil are spectrally mixed.
A09	Bare/sparse vegetation, Herbaceous wetland, Mangroves	Mid-generalist	Soil–flooding gradient	Gradient from dry bare soil to flooded herbaceous wetlands and mangrove systems, representing strong soil moisture and flooding dynamics across floodplains and coastal zones.
A12	Bare/sparse vegetation, Grassland, Shrubland	Mid-generalist	Fractional vegetation in drylands	Semi-arid landscapes characterized by gradual transitions from bare soil to sparse grasslands and shrublands where vegetation fraction varies continuously.
A13	Cropland, Grassland, Herbaceous wetland	Mid-generalist	Herbaceous vegetation systems	Land covers dominated by herbaceous vegetation including crops, natural grasslands and wetlands that share strong seasonality and similar canopy spectral signatures.
A36	Bare/sparse vegetation, Built-up, Moss/lichen	Mid-generalist	Highly reflective sparse vegetation	Reflective mineral or artificial substrates such as rock, bare soil and urban surfaces with minimal vegetation signals and strong brightness in visible bands.
A50	Cropland, Shrubland, Tree cover	Mid-generalist	Agroforestry mosaics	Agricultural landscapes where crops coexist with woody vegetation such as agroforestry systems, plantations or silvopastoral mosaics.
A51	Bare/sparse vegetation, Shrubland, Tree cover	Mid-generalist	Woody vegetation gradient	Gradual increase of woody vegetation from nearly bare surfaces to shrublands and open forests representing partial woody cover landscapes.
A55	Built-up, Moss/lichen, Snow/ice	Mid-generalist	High albedo surfaces	Surfaces with very high albedo and low chlorophyll signal such as snow, bright urban materials and rocks with lichens.
A59	Cropland, Grassland, Shrubland	Mid-generalist	Cropland, grassland, shrubland axis	Rural matrices where agricultural fields, fallows and semi-natural vegetation coexist forming a strong continuum of herbaceous and woody land covers.
A61	Grassland, Shrubland, Permanent water bodies	Mid-generalist	Riparian mosaics	Landscapes where open vegetation interacts with water bodies such as riparian corridors and piedmont areas combining vegetation reflectance and water absorption signatures.
A02	Cropland, Grassland, Shrubland, Snow/ice	High-generalist	Seasonal snow over open vegetation	Temperate and cold landscapes where open herbaceous vegetation periodically becomes snow covered, generating strong seasonal spectral changes.
A44	Bare/sparse vegetation, Grassland, Shrubland, Snow/ice	High-generalist	Arid–semiarid environments	Gradients of sparse vegetation typical of arid or semi-arid regions occasionally influenced by seasonal snow events.
A60	Bare/sparse vegetation, Cropland, Moss/lichen, Snow/ice	High-generalist	High latitude and mountain mosaics	Highly heterogeneous landscapes typical of high latitudes or mountains where bare soil, marginal croplands, lichens and snow coexist within extremely seasonal environments.

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land cover classes) are distinguished by their consistently low contribution to individual classification tasks. These dimensions have a more homogeneous distribution of importance across coverages and do not show a dominant association with any class.

This behavior suggests that these dimensions capture more general patterns, possibly associated with large-scale environmental gradients or variables that are not decisive for the discrimination of specific land cover types. This type of representation is consistent with the global nature of GAEF embeddings, which seek to provide a consistent description of the state of the Earth's surface on a planetary scale [11]. Although these dimensions do not contribute significantly to individual class discrimination, their presence suggests that robust land cover classification depends on the interplay between specialist and shared dimensions, not specialist dimensions alone.

E. Functional organization of the embedding space

The integration of the observed patterns reveals a functional structure in the embedding space, characterized by the coexistence of different types of dimensions according to their behavior in classification tasks. The results suggest a spectrum of functional roles: from specialist dimensions with high discriminatory capacity for a single land cover class, through low- and mid-generalist dimensions that capture relationships between two or three classes, to high-generalist dimensions that reflect broader environmental patterns.

This organization emerges consistently from empirical analysis, rather than from any explicit structural constraint known to the authors, suggesting that the embedding space encodes structured information related to the geographical organization of the territory. This finding contributes to advancing our understanding of foundation models, by showing that their representations are not arbitrary, but rather reflect patterns that can be interpreted from an ecological and spatial perspective.

F. Visual representation of the universe of embedding dimensions

The embedding universe visualization (Section 2.5) provides an integrated view of the identified patterns. Figure 6, generated using the interactive Dashboard [8], illustrates the embedding space at a 98% baseline performance recovery threshold. At this threshold, a predominance of specialist embedding dimensions is observed, with most dimensions positioned in close proximity to a single land cover class, suggesting a high capacity for specialization in the representation space. Shared dimensions, though fewer in number, occupy intermediate positions that highlight inter-class relationships. The visualization summarizes the associations and categories obtained in the preceding analyses. Its layout supports exploration of those results but does not provide an independent validation of the proposed functional interpretations.

G. Summary of findings

Overall, the results obtained do not seek to conclusively validate a hypothesis or to attribute definitive and unique interpretations to individual embedding dimensions, but rather to provide empirical evidence in support of a functional interpretability framework. The observation of consistent patterns in the importance of embeddings, their differential behavior between land cover classes, and their organization in the representation space suggests that it is possible to characterize these dimensions in functional terms — while acknowledging that the interpretations proposed here are task-dependent, biophysically informed approximations rather than fixed semantic labels.

Together, the results support a task-dependent taxonomy of recurring class–dimension associations. The workflow can be applied to other reference legends, regions, or representations, but the persistence of the reported roles under those changes remains an empirical question.

IV. Discussion

A. Interpretability in GFMs: from physical variables to functional structures

GFMs have demonstrated significant progress in Earth observation tasks, thanks to their ability to integrate multi-source data into latent representations of high dimensionality [4, 11]. However, this progress has been accompanied by a fundamental challenge: the lack of interpretability of embeddings, which encode environmental information into abstract vectors with no direct correspondence to observable physical variables [17].

Existing approaches have addressed this limitation primarily by associating embedding dimensions with continuous variables, such as temperature, precipitation, or elevation, seeking to identify explicit physical relationships within the latent space [17]. While these approaches represent an important advance, they focus on a physical-variable interpretation, which does not necessarily capture the categorical organization of the territory in terms of land cover.

In contrast, this paper proposes a complementary approach based on functional interpretability, in which embeddings are interpreted according to their role in discriminating between coverages and representing their interactions. This approach allows the latent space to be connected to fundamental geographic concepts, such as coverage units and transition zones, extending the scope of interpretability beyond traditional physical variables.

B. Functional organization of the embedding space

The results suggest that the embedding space has a non-uniform structure, where dimensions play different roles. This organization can be interpreted as a functional spectrum, from specialist dimensions that capture the distinctive characteristics of a single land cover class, through low- and mid-generalist dimensions that encode relationships between two or three classes, to high-generalist dimensions that reflect broader environmental patterns.

Class-specific and shared associations suggest hypotheses about how land-cover information is distributed across the released coordinates. One possibility is that some dimensions respond preferentially to relatively homogeneous class examples, whereas others respond to properties shared across classes.

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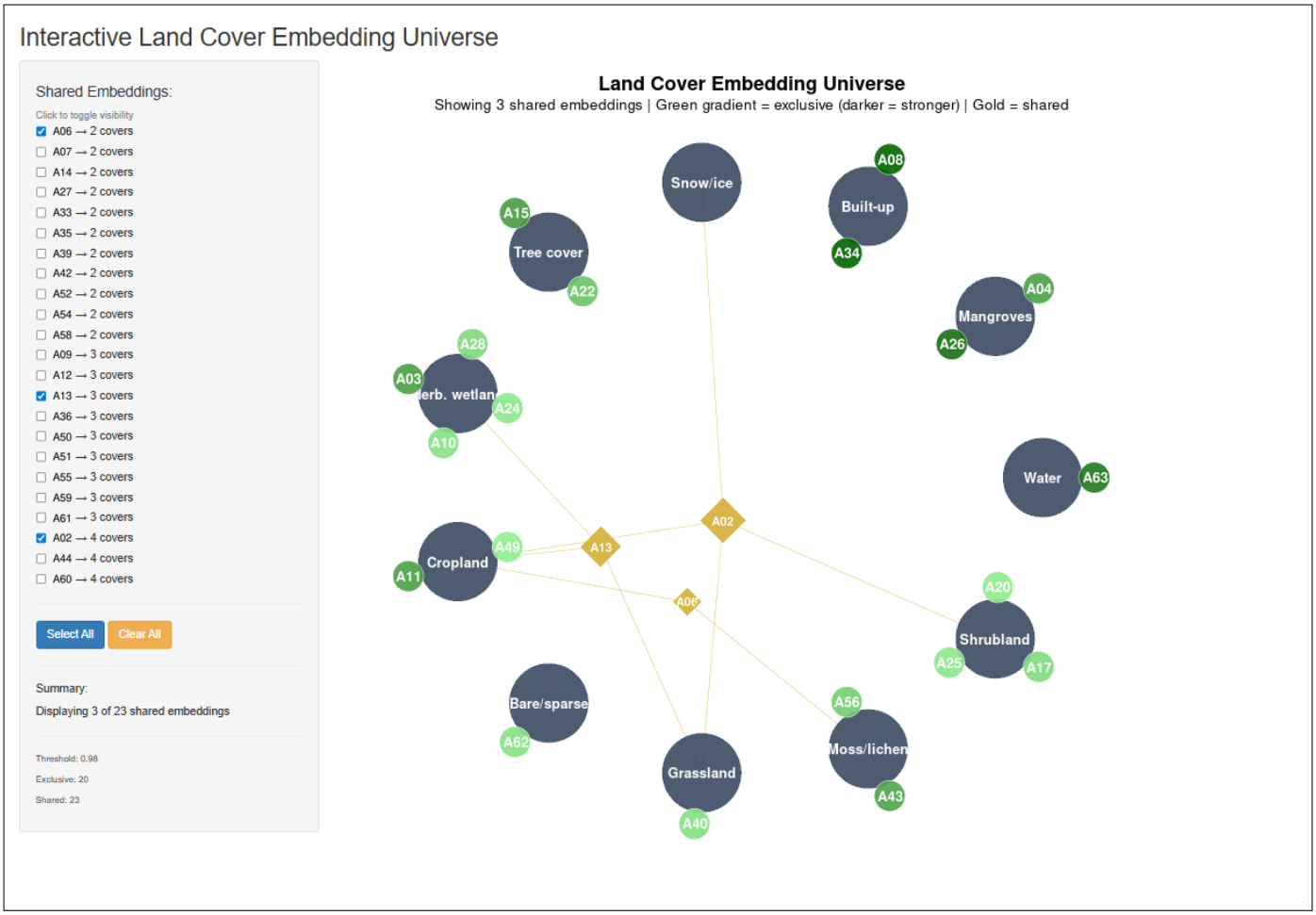


Fig. 6: "Embedding Universe" visualization at a 98% baseline performance recovery threshold, generated using the interactive Dashboard [8]. Green nodes represent specialist dimensions (darker = stronger association); gold nodes represent shared dimensions. See Section 2.5 for a full description of the visual encoding. To interact with this plot, access this link. Node positions are assigned for visualization and do not represent measured distances in the original embedding space or geographic locations of ecological transitions.

Testing this explanation requires explicit comparisons between class interiors, boundaries, and mixed landscapes. The present classification experiments do not distinguish these mechanisms.

This functional structure emerges consistently from empirical analysis, suggesting that embeddings are not arbitrary representations but rather reflect regularities inherent in the organization of the territory. In this sense, the embedding space can be understood as a system in which different dimensions capture different levels of abstraction of geographical reality.

This finding is consistent with evidence that foundation models are capable of learning rich and generalizable representations from heterogeneous data [4], but it offers a new perspective by suggesting that these representations also possess a structure that can be interpreted in terms of discrete spatial categories and their interactions.

C. Spatial consistency of interpreted dimensions

Applying the spatial-consistency index to all interpreted specialist dimensions revealed a clear gradient in how uniformly each dimension discriminates its associated class across the Earth's surface (Figure 7). The permanent water dimension (A63) achieved the highest consistency, confirming that its association with water is not a regionally localized effect but a spatially uniform property maintained across the full range of sampled locations. This makes A63 the clearest example of a globally stable specialist: the class it encodes is both spectrally distinctive and geographically ubiquitous, and its discriminative role does not weaken in any region.

Most remaining specialists occupied an intermediate range, indicating dimensions that are broadly informative but exhibit some spatial modulation in their discriminative strength. Notably, for classes represented by more than one exclusive dimension, the paired dimensions did not behave identically: within Tree cover and Built-up, one member of each pair scored consistently higher than the other, indicating that the two specialists of a single class are not redundant but capture spatially distinct expressions of that class. This is consistent with the interpretation that broadly distributed land cover types spanning multiple biomes are encoded by complementary dimensions specialized to different geographic regimes, rather

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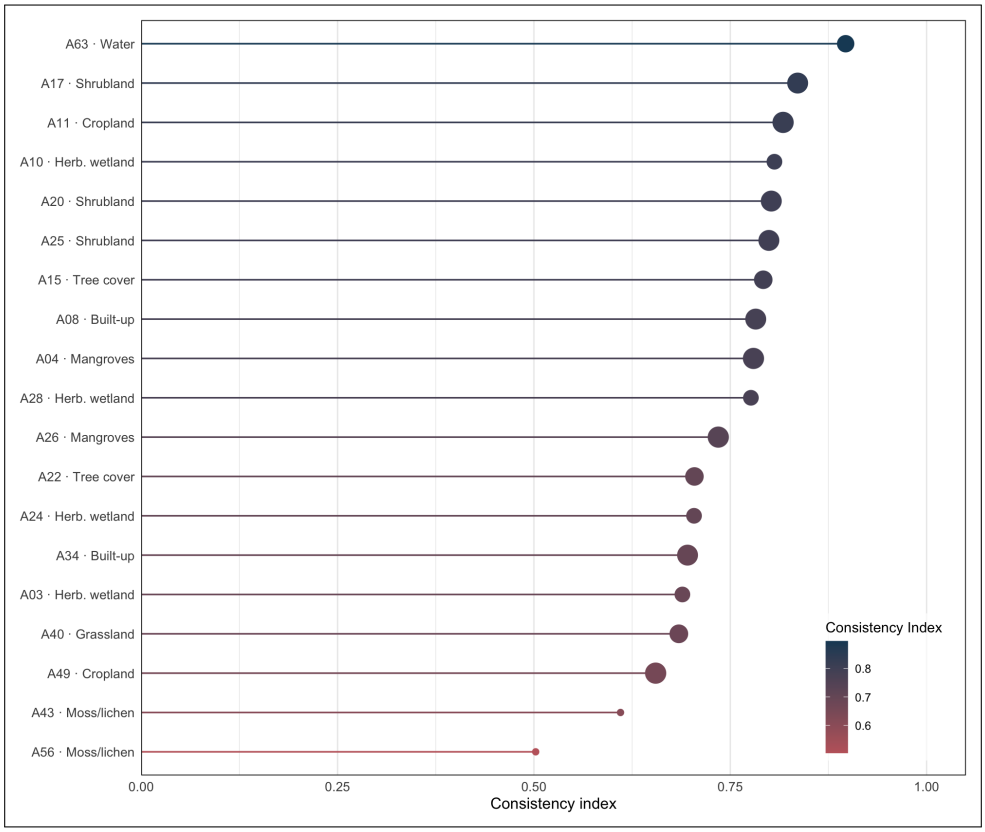


Fig. 7: Spatial consistency of interpreted specialist embedding dimensions. Each dimension is scored by the spatial uniformity of its top-two frequency across a $50^\circ \times 50^\circ$ longitude $\times$ latitude grid (cells with fewer than 15 experiments excluded), where 1 indicates a dimension that discriminates its associated class uniformly wherever the class is sampled and 0 indicates strongly localized behavior. Point size is proportional to the total number of experiments underlying each score. The permanent water dimension (A63) is the most spatially uniform specialist, whereas the moss/lichen dimensions (A43, A56) are the least; classes represented by paired exclusive dimensions (Tree cover, Built-up) show the two members ranking apart, indicating geographically complementary specialization.

than by a single globally uniform dimension.

At the low end of the gradient, the moss/lichen dimensions (A43, A56) showed the weakest spatial consistency, in line with their restriction to high-latitude cryogenic environments. It should be noted that the index measures uniformity across the regions where a class is actually sampled, so geographically restricted classes are assessed for consistency within their natural range rather than across the entire globe; a high score therefore denotes uniformity where the class occurs, not planetary ubiquity. Taken together, these results indicate that the functional roles identified in the preceding sections are, for most dimensions, spatially robust rather than artefacts of particular regions, while also revealing that a class's specialist dimensions can partition its representation along geographic lines.

D. Class-specific associations and hypotheses about class interiors

The identification of embeddings with high specialization in certain coverages suggests that the latent space contains dimensions that capture distinctive characteristics show recurring discriminative relevance of specific categories of the territory. This behavior is particularly evident in coverages with well-defined physical signatures, where the separability between classes is clearer.

From an interpretative perspective, these embeddings can be understood as representations of the "cores" of land cover classes, i.e., regions where the characteristics of a class are dominant and relatively homogeneous. In these contexts, specialized embeddings enable accurate discrimination, acting as robust indicators of the presence of a land cover class.

This type of representation is consistent with the ability of foundation models to capture relevant physical properties of the environment, although in this case the interpretation is performed in terms of geographic categories rather than continuous variables [17].

E. Shared associations and hypotheses about transition zones

One of the most significant contributions of this work is the identification of generalist embedding dimensions associated with a limited number of land cover classes. These dimensions serve a dual role: they capture characteristics common to classes that overlap spatially or functionally, as in transition zones or ecotones where coverages lack clearly defined boundaries, and

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they complement specialist dimensions in the classification of individual land cover classes where no single dimension is sufficient on its own.

The existence of these dimensions suggests that foundation models not only capture discrete patterns, but also the interactions between them, integrating information from multiple sources and spatial scales. This behavior is consistent with the nature of environmental systems, where transitions between land covers are frequent and play a key role in the dynamics of the territory. Their association with two or three specific classes makes low- and mid-generalist dimensions particularly suited to representing the inter-class relationships characteristic of ecotonal boundaries.

In this sense, these dimensions constitute a bridge between the continuous representation of latent space and the discrete structure of coverages, allowing the complexity inherent in geographic systems to be captured. This aspect has been little explored in the literature, despite its relevance for applications such as monitoring land use changes or detecting degradation processes.

F. Broad class associations and candidate environmental explanations

In addition to specialist, low-, and mid-generalist dimensions, high-generalist dimensions are identified that exhibit more general behavior, without a dominant association with any individual land cover class. These dimensions can be interpreted as representations of large-scale environmental gradients, such as climatic or topographic conditions.

This type of dimension reflects the global nature of GFMs, which seek to provide a consistent representation of the planet across different regions and contexts [11]. Although these dimensions are not highly discriminative in classification tasks, their presence indicates that the embedding space incorporates information relevant to the general characterization of the environment.

From an interpretive perspective, high-generalist dimensions can be considered as a higher level of abstraction, capturing properties of the system that transcend the discrete categories of land cover.

G. Uninterpreted dimensions and their significance

Of the 64 embedding dimensions in GAEF's latent vector space, the preceding analyses yield functional interpretations for 43. The remaining 21 dimensions were not selected for any of the aggregate class-level interpretive sets. These uninterpreted dimensions may encode information outside the land cover domain — such as broader environmental gradients, atmospheric conditions, or properties of the built environment — or may represent aspects of the Earth's surface that the current experimental framework, focused on land cover classification, is not designed to capture.

This constitutes an important direction for future research. Cross-domain experiments targeting alternative domains (e.g., infrastructure, climate, or land use) could reveal whether these dimensions carry structured information that becomes legible under a different contextual lens. Additionally, targeted validation experiments using other datasets may help determine whether these dimensions truly encode subtle environmental signals that fall below the discrimination threshold in the current design.

H. Implications for interpretability and remote sensing

The proposed framework has important implications for the interpretability of foundation models in remote sensing. First, it allows interpretation to shift from a variable-based approach to a function-based approach, which is more suitable for analyzing complex geographic phenomena.

Secondly, identifying different types of embedding dimensions opens up the possibility of designing more efficient and explainable models by selecting subsets of dimensions according to their function. For example, specialist dimensions could be used to improve accuracy in class-specific identification, while low- and mid-generalist dimensions could be key to improving change detection or edge delimitation.

Thirdly, this approach allows the interpretation of models to be integrated with concepts specific to geography and ecology, facilitating their use in scientific and decision-making contexts.

I. Practical implications

The framework can support exploratory feature prioritization and the design of targeted validation experiments. The reported class–dimension associations provide candidate subsets for evaluation in specific applications. Operational use requires confirmation of predictive performance, computational savings, and spatial or temporal transferability in the intended setting. Applications to ecotone monitoring, change detection, and early warning remain prospective.

J. Limitations of the study

Despite the contributions presented, this work has several limitations that should be considered. First, the analysis is based on importance measures derived from machine learning models, which may be influenced by the algorithm used and the distribution of the data. This implies that the interpretation of the embeddings depends, to some extent, on the experimental context.

The analysis concerns a single pretrained representation and a single reference year. Random training–test splits within individual ROIs assess discrimination in the sampled contexts, rather than transfer to spatially held-out regions. Moreover, the functional labels describe behavior in the released coordinate system; their stability under alternative representations, model versions, or coordinate transformations has not been established. Class-level interpretive sets assembled from aggregate top-two frequencies were not evaluated as fixed predictor sets. The performance-recovery results concern subsets selected separately within each experiment and therefore do not establish equivalent performance for the aggregate sets. Their predictive performance requires separate validation before they can be recommended for operational feature selection.

The functional classification also depends on the selected relative performance-recovery threshold of 0.98. Changing

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this threshold can alter the class-level dimension counts and, consequently, the composition of the interpretive sets and the assigned specialist–generalist categories. The robustness of these assignments to the threshold choice remains to be assessed.

Additionally, although the proposed visualization facilitates interpretation, it represents a simplification of the embedding space, which is actually high-dimensional. Therefore, the relationships observed should be understood as an approximation of the actual structure of the system.

Furthermore, although this study captures regions of interest distributed globally, the functional roles identified for each embedding dimension may shift when the framework is applied at a regional scale. Classification accuracy itself varies geographically (Supplementary Figure 2), suggesting that regional land cover conditions influence the discriminatory capacity of the embedding space. In geographically constrained contexts, reduced land cover diversity and environmental variability could diminish the discriminatory capacity of dimensions that are informative at a global level, effectively flattening the importance distribution.

Finally, this study does not address the validation of the proposed interpretations in terms of specific physical variables or environmental processes, which constitutes a line of future work.

K. Future lines of research

The results obtained open up multiple lines of research. One relevant direction is the validation of the proposed framework by associating embeddings with specific physical variables, which would allow the integration of functional and physical interpretability approaches.

Another line of research consists of analyzing the stability of the functional classification of embeddings in different regions and temporal scales, evaluating their capacity for generalization. Likewise, the study of the temporal dynamics of embeddings could provide information on processes of change in the territory.

A third line of research revolves around geographic variation. The geographic variation in classification performance revealed across the global experiments (Supplementary Figure 2) suggests that the functional roles of individual embedding dimensions may not be uniform across biomes or climatic zones. A systematic investigation of the geographic determinants of embedding discriminatory capacity constitutes a promising direction for future research.

Additionally, the proposed framework could be extended to other remote sensing tasks, such as change detection, semantic segmentation, or environmental variable prediction, exploring the role of different types of embeddings in each context.

Finally, integrating these results into interactive analysis and decision-making systems — such as climate adaptation planning tools and environmental monitoring platforms — represents an opportunity to bring foundation models closer to practical applications, facilitating their interpretation by non-specialist users.

V. Conclusion

This study presents a post hoc framework for exploring the functional roles of AlphaEarth Foundations embedding dimensions in land-cover discrimination. By combining geographically distributed classification experiments, feature-ranking summaries, reduced-dimension performance profiles, and spatial analysis, it characterizes recurring class-specific and shared associations within the evaluated tasks.

The main contribution is an operational approach to organizing these associations and formulating testable interpretations. The proposed specialist–generalist taxonomy describes association breadth under the study design. It does not establish unique physical meanings for individual dimensions or an intrinsic hierarchy within the foundation model.

The findings motivate further evaluation of task-specific dimensionality reduction and targeted investigation of the environmental properties associated with selected dimensions. Validation across independent regions, years, reference datasets, and pretrained representations is needed to establish the scope of these interpretations and their value for operational applications.

VI. Statements and Declarations

A. Competing Interests

The authors have no competing interests to declare that are relevant to the content of this article.

B. Clinical trial number

Not applicable.

C. Funding

The authors did not receive support from any organization for the submitted work.

D. Ethical responsibilities of Authors

All authors have read, understood, and have complied as applicable with the statement on “Ethical Responsibilities of Authors” as found in the Instructions for Authors.

E. Data Availability Statement

No new primary data were created in this study. The analysis was conducted using publicly available remote sensing data from the Copernicus Open Access Hub. The derived data products supporting the findings of this study are available from the corresponding author upon request.

F. Code Availability Statement

The source code for the “What on Earth is AlphaEarth?” interactive Dashboard is available at [7]. The experimental notebook used to conduct the classification experiments is available at [2].

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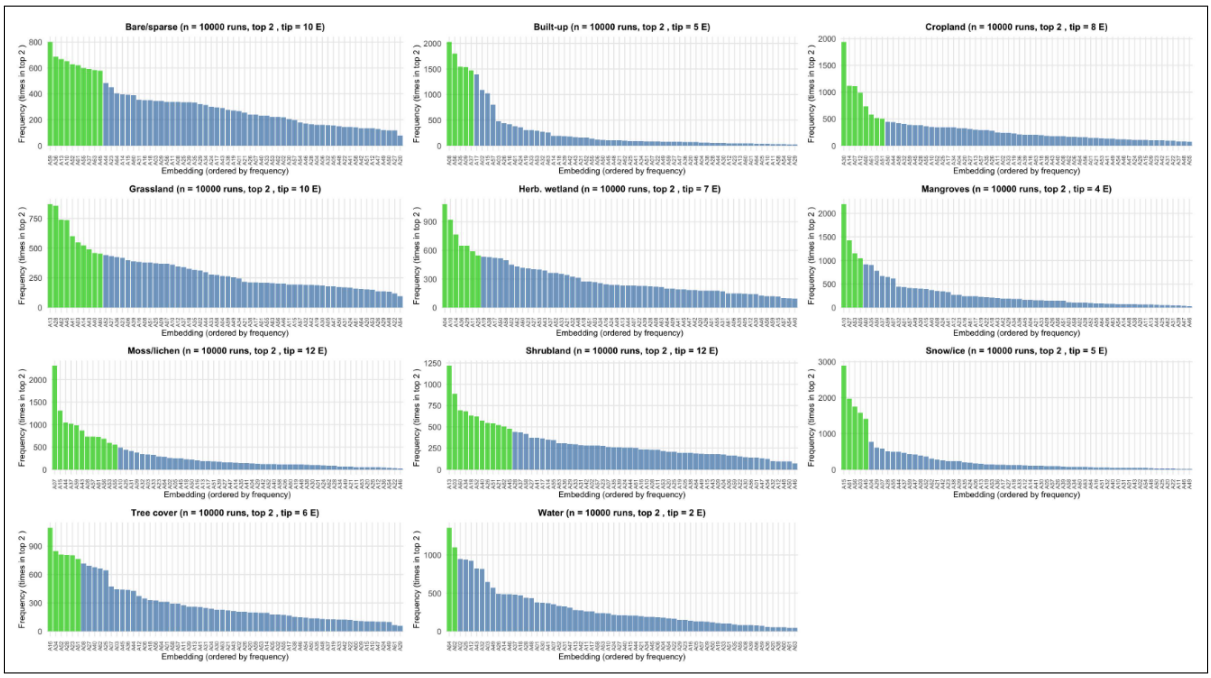
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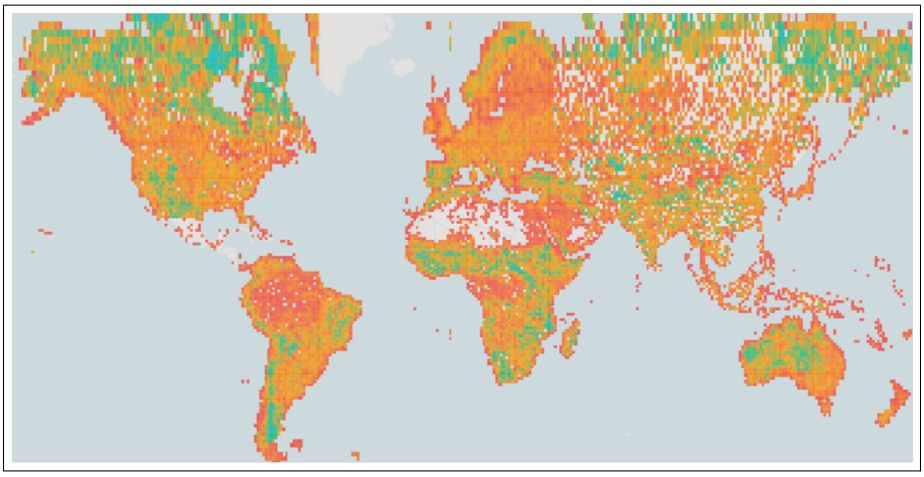
Supplementary Information

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Supplementary Figure 1: Embedding frequency distributions by land cover class. For each class, embedding dimensions are ranked by frequency of appearance in the top 2 most important positions across 10,000 experimental runs. Green bars indicate dimensions within the tipping point threshold; blue bars indicate remaining dimensions. Generated using the interactive Dashboard [8].



Supplementary Figure 2: Grid-based geographic performance heatmap of classification accuracy across global experiments. Each cell represents the mean classification accuracy aggregated across all experiments conducted within that geographic grid cell. Red cells indicate higher accuracy ($> 90\%$), orange cells indicate moderate accuracy (80–90%), and blue cells indicate lower accuracy ($< 80\%$). Generated using the interactive Dashboard [8].

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